1. Efficient Market Hypothesis (20%)
    a) Please explain the three different versions of the efficient market hypothesis (EMH). (5%)
    b) Please provide two empirical pieces of evidence against weak EMH. (5%)
    c) Please provide empirical evidence against the semi-strong version of EMH, which is different from the two pieces of evidence you provide in (b) (5%)
    d) What are the limits to arbitrage argument and why it is crucial for behavioral finance schools to fight against the critics of the rational school? (5%)

2. Stock market anomalies (15%)
    a) Please describe one stock market "anomaly", provide an explanation from both rational (EMH) school and behavioral school, and develop a strategy to exploit the anomaly.

3. Valuation (25%)
    a) Please explain what DCF and relative pricing (please provide the main steps) is (10%)
    b) Please compare the advantages and disadvantages of DCF and relative pricing (10%)
    c) Please discuss whether paying dividends would help boost firm valuation (5%)

4. APT (20%)
    In a two-factor financial market, we have known the realized return of three assets *A*, *B*, and *C* as follows:

$$\begin{aligned}\tilde{r}_A &= 0.04 + 2\tilde{F}_1 - 2\tilde{F}_2 + \tilde{\epsilon}_A \\ \tilde{r}_B &= 0.03 - 1\tilde{F}_1 + 2\tilde{F}_2 + \tilde{\epsilon}_B \\ \tilde{r}_C &= 0.11 + 3\tilde{F}_1 + \tilde{\epsilon}_C\end{aligned}$$

If an asset D has a loading of 4 on the factor F_1 and a loading of 3 on the factor F_2, please use two methods to calculate the mean return of D. (10%)

5. Fund Managers (10%)
    a) Do you think fund managers can outperform the stock market index return in China? Please provide your thoughts and evidence.

6. Behavioral Finance (10%)
    a) Please describe two behavioral biases of individual investors in the stock market. Please discuss the phenomenon, explain why it is a bias, and provide the psychological explanation for the bias.